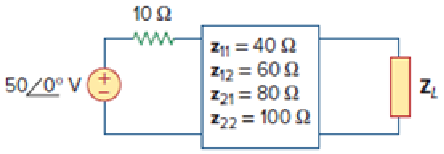
Chapter 19, Problem 13P

Problem

Determine the average power delivered to *ZL* = 5 + *j*4 in the network of Fig. 19.74. *Note:* The voltage is rms.

Figure 19.74



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

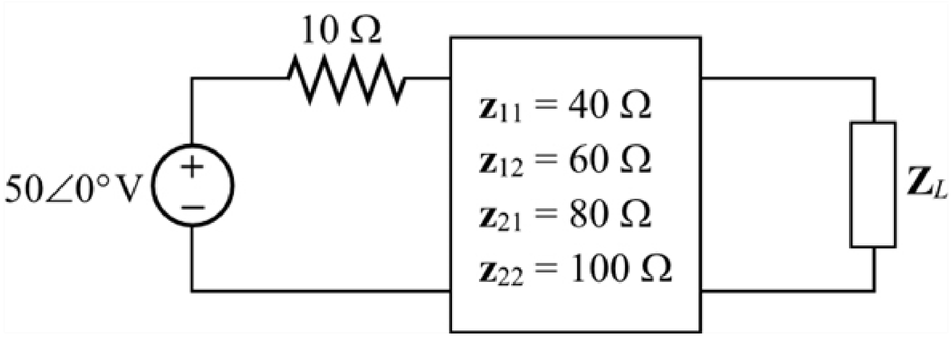


Figure 1

Step 2 of 7

Write the current and voltage expressions of $\mathbf{z}$ parameters.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$

$$= 40\mathbf{I}_1 + 60\mathbf{I}_2 \;\cdots\cdots (1)$$

And,

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

$$= 80\mathbf{I}_1 + 100\mathbf{I}_2 \;\cdots\cdots (2)$$

Step 3 of 7

Draw the circuit with voltage notations and current directions.

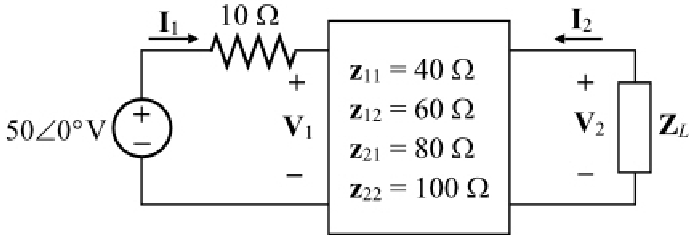


Figure 2

Step 4 of 7

Apply Kirchhoff's voltage law to the right side loop.

$$10\mathbf{I}_1 + \mathbf{V}_1 = 50\angle 0°$$

$$\mathbf{V}_1 = 50\angle 0° - 10\mathbf{I}_1 \quad \text{......} \ (3)$$

From Figure 2, the voltage across the load impedance:

$$\mathbf{V}_2 = -\mathbf{Z}_L\mathbf{I}_2$$
$$= -(5 + j4)\mathbf{I}_2 \quad \text{......} \ (4)$$

Step 5 of 7

Substitute $50\angle 0° - 10\mathbf{I}_1$ for $\mathbf{V}_1$ in the equation (1).

$$50 - 10\mathbf{I}_1 = 40\mathbf{I}_1 + 60\mathbf{I}_2$$
$$50\mathbf{I}_1 + 60\mathbf{I}_2 = 50$$

$$5\mathbf{I}_1 + 6\mathbf{I}_2 = 5 \quad \text{......} \ (5)$$

Substitute $-(5 + j4)\mathbf{I}_2$ for $\mathbf{V}_2$ in the equation (1).

$$-(5 + j4)\mathbf{I}_2 = 80\mathbf{I}_1 + 100\mathbf{I}_2$$
$$80\mathbf{I}_1 = -100\mathbf{I}_2 - (5 + j4)\mathbf{I}_2$$
$$\mathbf{I}_1 = -\left(\frac{105 + j4}{80}\right)\mathbf{I}_2$$

$$= -(1.3125 + j0.05)\mathbf{I}_2 \quad \text{......} \ (6)$$

Step 6 of 7

Substitute $-(1.3125 + j0.05)\mathbf{I}_2$ for $\mathbf{I}_1$ in the equation (5).

$$-(5)(1.3125 + j0.05)\mathbf{I}_2 + 6\mathbf{I}_2 = 5$$
$$(-6.5625 - j0.25 + 6)\mathbf{I}_2 = 5$$
$$(0.5625 - j0.25)\mathbf{I}_2 = 5$$

$$\mathbf{I}_2 = \frac{5}{0.5625 - j0.25}$$
$$= -7.4226 + j3.2989$$
$$= 8.123\angle 156°\ \text{A}$$

Step 7 of 7

Calculate the average power delivered to the load.

$$
\begin{aligned}
P &= |\mathbf{I}_2|^2 Z_L \\
&= (8.123)^2 (5) \\
&= (65.98)(5) \\
&= 330 \text{ W}
\end{aligned}
$$

Therefore, the average power, P delivered to the load is $\boxed{330 \text{ W}}$.